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K/U	A	T	C	Total
19	8	11	11	49

SCH4U Structure & Properties Unit 2 Test

Multiple Choice (K/U)

Identify the letter of the choice that best completes the statement or answers the question.

- _____ 1. The 3p atomic orbital has the shape of
 - a. a sphere
 - b. a torus
 - c. a dumb-bell
 - d. two perpendicular dumb-bells
 - e. an egg
- _____ 2. How does atomic radius change from left to right across a period of the Periodic Table?
 - a. it tends to decrease
 - b. it tends to increase
 - c. it does not change
 - d. it first increases, and then decreases
 - e. it first decreases and then it increases
- _____ 3. Why do energy levels exist in atoms?
 - a. electrons are negatively charged
 - b. electrons are attracted to certain numbers of neutrons
 - c. electrons are able to possess any range of energy
 - d. electrons will only display certain colours
 - e. electrons are only able to possess quanta of energy
- _____ 4. What was Planck's contribution to the quantum mechanical model of the atom?
 - a. the uncertainty principle
 - b. the concept of quanta of energy
 - c. the idea that every mass has a wave with which it is associated
 - d. the wave equation
 - e. a relationship between energy and mass
- _____ 5. What did Schrodinger contribute to the quantum mechanical model of the atom?
 - a. the uncertainty principle
 - b. concept of quanta of energy
 - c. the idea that every mass has a wave with which it is associated
 - d. the wave equation
 - e. a relationship between energy and mass
- _____ 6. Which of the following is the electron configuration for neon?
 - a. $1s^2 2s^2 2p^4 3s^2$
 - b. $1s^1 2s^1 2p^6 3s^2$
 - c. $1s^2 2s^2 2p^6$
 - d. $1s^3 2s^3 2p^4$
 - e. $1s^2 2s^2 2p^8$
- _____ 7. What made Rutherford believe that atoms are mostly empty space?
 - a. most alpha particles went straight through the gold foil
 - b. some alpha particles were deflected by the gold foil
 - c. the line spectra of excited atoms
 - d. atoms are electrically neutral
 - e. none of the above
- _____ 8. Bohr's model of the atom can be termed a "quantum" model of the atom because
 - a. electrons and protons are equal
 - b. it is based on the solar system
 - c. the nucleus is positive
 - d. each electron can possess only specific amounts of energy
 - e. the potential energy of electrons is measurable

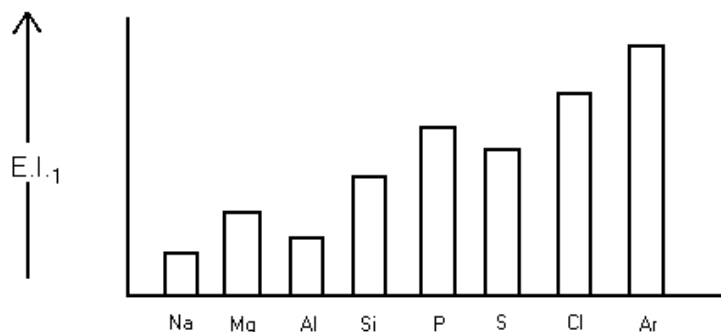
9. Which of the atoms could have the following valence electron configuration?
- | | | | |
|----------------------|------------|----------------------|------------|
| $\uparrow\downarrow$ | $\uparrow$ | $\uparrow\downarrow$ | $\uparrow$ |
| s | | p | |
- a. carbon
b. nitrogen
c. boron
d. sulfur
e. none of the above
10. Which of the following is the configuration of carbon?
- a. $1s^2 2s^2 2p^2$
b. $1s^2 2s^2 2p^4$
c. $1s^2 2s^1 2p^3$
d. $1s^2 2s^1 2p^4$
e. none of the above
11. Polar covalent bonds occur between
- a. atoms which both have equally high electronegativities
b. atoms which have high but unequal electronegativities
c. atoms which both have equally low electronegativities
d. atoms which both have equally low ionization energies
e. atoms which have low but unequal ionization energies
12. Which of the molecules, CO_2 , H_2O , NH_3 , and BF_3 , will be polar?
- a. CO_2 , NH_3 and BF_3
b. H_2O and NH_3
c. H_2O and BF_3
d. CO_2 , H_2O and NH_3
e. CO_2 and BF_3
13. Why does a central atom surrounded by 4 atoms have a tetrahedral shape instead of a square planar shape?
- a. orbitals are never at right angles to each other
b. the lone pairs around the central atom push the 4 atoms into this configuration
c. the angle in a tetrahedron is larger than in a square planar arrangement
d. the 4 atoms want to be as close together as possible
e. none of the above
14. Which forces exist between iron, Fe, particles?
- I. Van der Waals
II. metallic bonding
III. hydrogen bonding
IV. dipole
- a. I only
b. I and IV only
c. I and II only
d. I, III and IV only
e. I, II and III only
15. Which statement is the best description of chlorine, Cl_2 ?
- a. polar molecule
b. polar bonds, non polar molecule
c. non polar molecule
d. ionic compound
e. none of the above
16. Which statement is the best description of potassium chloride, KCl ?
- a. polar molecule
b. polar bonds, non polar molecule
c. non polar molecule
d. ionic compound
e. none of the above
17. What type of substance is carbon dioxide, CO_2 ?
- a. ionic
b. molecular
c. covalent network
d. metallic
e. none of the above
18. What type of substance is hydrogen sulfide, H_2S ?
- a. ionic
b. molecular
c. covalent network
d. metallic
e. none of the above

c. covalent network

- _____ 19. How will four pairs of electrons surrounding a central atom be arranged?
- | | |
|------------------|---------------|
| a. pyramidally | d. linearly |
| b. spherically | e. trigonally |
| c. tetrahedrally | |

Short Answer (I/C)

20. Ionization energy tends to increase from left to right on the periodic table. There are two exceptions to that rule in the graph below. Explain them using Quantum Theory. (2 marks)



21. Draw and write the electron configuration for calcium. (2 marks)
22. Draw the electron configuration for oxygen and explain how you use Hund's Rule and the Pauli Exclusion principle to do it. (2 marks)
23. Why is sulfur able to bond with six different atoms at once? (2 marks)
24. Distinguish between single and double bonds with respect to the bonding orbitals. (2 marks)
25. Draw the electron configuration for *Scandium, Sc*. (2 marks)
26. What is meant by the term "orbital"? (2 marks)

27. Explain how VSEPR theory explains the shape of a central atom surrounded by two unbonded pairs of electrons and two other atoms. (2 marks)
28. Predict the shape of PF_5 . Indicate the bond angles. (2 marks)
29. Predict the shape and bond angle(s) of beryllium difluoride, BeF_2 . (2 marks)
30. Why do metals conduct electricity? (2 marks)

Essay (MC) Answer only ONE of the following questions. (8 marks)

31. Explain water's properties using what you know about its inter- and intramolecular forces.
32. Explain VSEPR theory in your own words. Use examples.